MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Core Course (CC) - MG4CCRBCA200 - ARTIFICIAL INTELLIGENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

VALUATION SCHEME

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions, Each question carries **2** marks

1. **Identify the primary goal of a problem-solving agent for a problem scenario.** [K] / [CO1]
   Answer: Primary goal is to identify the initial state, final state, find path to the final state, estimation of cost to reach the final state and to choose the solution with minimum cost.
   Mark distribution: Any answer with related meaning may be given 2 marks.

2. **What is an AND-OR graph?** [K] / [CO2]
   Answer: It is an extension of the A* algorithm; two types of nodes- AND nodes represent tasks where all child nodes must be satisfied, **OR** nodes with multiple alternative paths
   Mark distribution: Any answer with related meaning may be given 2 marks.

3. **What is meant by optimality in artificial intelligence search techniques?** [K] / [CO2]
   A search algorithm is said to be optimal if it always returns a solution with the lowest cost among all possible solutions. Major terms: low cost, least time, maximum efficient output.
   Mark distribution: any answer with related meaning may be given 2 marks

4. **Explain the procedure of forward chaining.** [U] / [CO3]
   Starts with a set of known facts and applies **IF–THEN rules** to infer new facts. The process begins by checking the current facts with the conditions of available rules; when a rule's conditions are satisfied, its conclusion is added to the knowledge base.
   Mark distribution: Any answer with related meaning may be given 2 marks.

5. **What is meant by Traditional rule based expert system?** [U] / [CO4]
   Traditional Systems depend on logical if-then rules stored in the KB to make decisions or draw conclusions. This system will generate a response when the

query is related to the stored knowledge or facts. Otherwise, it will fail.
Mark distribution: Any answer with related meaning may be given 2 marks.

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions, Each question carries **6** marks

6. **Explain Wumpus World problem in detail.** [U] / [CO1]

The Wumpus computer game, The agent explores a cave consisting of rooms connected by passageways, somewhere in the cave is the Wumpus, a beast that eats any agent that enters its room. Some rooms contain bottomless pits that trap any agent that wanders into the room. The Wumpus can fall into a pit too, so avoids them. Occasionally, there is a heap of gold in a room.The goal is to collect the gold and exit the world without being eaten.

Major terms: **Pits**: **breeze, Wumpus, stench, arrow**

PEAS: **Performance measures:** (Agent gets gold and return back safe = *+1000 points,* Agent dies (pit or Wumpus)= *-1000 points,* Each move of the agent = *-1 point,* Agent uses the arrow = *-10 points)* **Environment:(** A cave with *16(4x4)* rooms)**Actuators:** (Move forward,,Turn right/left: Rotate Shoot: Grab, Drop)**Sensors:** (Breeze,Stench, Glitter,Scream,Bump)

Mark Distribution: Major terms(2 marks), PEAS description(2 marks), General overview of the game with or without figure(2 marks)

7. **Describe the different nature of environments in Artificial Intelligence.** [U] / [CO1]

1. Fully observable(chess board) vs Partially Observable(card's game)
2. Static(crossword puzzle) vs Dynamic(dynamically challenging-driving)
3. Discrete(2x+3y=18) vs Continuous(driving)
4. Deterministic(next=current state+ action) vs Stochastic(rolling a die)
5. Single-agent(single player-crossword) vs Multi-agent(two-agent chess)
6. Episodic(next episode is independent) vs sequential(current episode determines next episode)
7. Known(all rules are known) vs Unknown()
8. Accessible(all info are accessible) vs Inaccessible

Mark Distribution: Any 6 may be given with 1 mark each

8. **Explain the concept of Rationality in AI. What makes an agent rational?** [U] / [CO1]

A rational agent is said to perform the right things. Rational agents are very similar to intelligent agents. Rational agent is capable of taking best possible action in any situation.

In artificial intelligence, rationality refers to an agent's ability to choose actions that maximize its performance measure, based on what it knows and perceives. A rational agent is one that always tries to do the right thing—that is, take the action expected to produce the best outcome given the situation.

Marks: Any answer with related meaning may be given marks.

9. **Analyze the impact of heuristics and constraint propagation methods on the structure and complexity of CSP solving.** [An] / [CO2]

Mark Distribution : About CSP- (major terms includes variables, domain, constraints, backtracking)(4 marks), Selection of heuristics- 2 marks

10. **Examine and interpret the roles of selection, crossover, and mutation in Genetic Algorithms in achieving evolutionary optimization.** [U] / [CO2]

Basic terms: selection, crossover, and mutation.
Selection- Selecting the individual solutions based on fitness function.
Crossover combines genetic information from two parent individuals to produce offspring.
Mutation randomly alters one or more genes in an individual.Prevents the algorithm from getting stuck in local optima.
Mark Distribution: Basic idea of GA(2 marks), Major terms(2 marks), diagram or example(2 marks)

11. **List the differences between forward chaining and backward chaining.** [K] / [CO3]

Forward chaining is a data-driven approach that starts with known facts and applies IF–THEN rules to generate new facts until a goal is reached; it moves step by step from available information toward conclusions.
In backward chaining is a goal-driven approach that starts with a desired goal and works backward by checking which rules could produce that goal, then verifying whether their conditions are true (or can be proven).
Mark distribution : Forward chaining with diagram or example- 2 marks, Backward chaining with diagram or example-2 marks, Difference- 2 marks

12. **Explain the concept of unsupervised learning.** [U] / [CO4]

Unsupervised learning is a type of machine learning where a model learns patterns and structures from unlabeled data. The system must discover hidden relationships on its own.
Key terms/ characteristics- unlabeled dataset, pattern discovery, exploratory data analysis, eg: clustering.
Applications: image pattern recognition, anomaly detection, customer segmentation

Mark Distribution: Major concept of unsupervised learning- 3 marks, example- 1 mark, applications- 2 marks

[6x5 = 30]

Part C

Essay Questions

Answer any **2** questions, Each question carries **15** marks

13. **Describe MIN-MAX algorithm in detail with its principle, algorithm and methods to enhance performance along with an example.** [A] / [CO2]

Mark distribution : MIN-MAX algorithm or diagrammatic explanation-8 marks, Drawback of MIN-MAX algorithm -3 mark, Alpha Beta Pruning-4 mark

14. **Explain the working of inference mechanisms in First Order Logic.** [U] / [CO3]

Mark distribution: First Order Predicate logic- 3 mark, Resolution- 6 marks, Forward chaining and backward chaining(3 marks each)

15. **What is meant by NLP? Explain the different core NLP Components?** [U] / [CO4]

NLP Basics- 5 marks, Core components(Speech recognition, Morphological Analysis, Syntactic Analysis, Semantic Analysis, Pragmatics, Discourse Analysis, Machine Translation etc) (Any 5 terms with 2 marks each)

[15x2 = 30]

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